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Ageing and hair cycles

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Summary The phototrichogram is a non-invasive technique by which, on the same precise area of the scalp, each individual hair may be identified, and its current growth phase established. This technique was used to study the duration of hair cycles in 10 male subjects, balding and non-balding, by observations at monthly intervals over a period of 8–14 years. The accumulated data served to characterize the effects of ageing in these subjects: a reduction in the duration of hair growth and in the diameter of hair shafts, most evident in the thickest hairs, and a prolongation of the interval separating the loss of a hair in telogen and the emergence of a replacement hair in anagen. These various aspects of ageing of scalp hair contribute to its progressive overall impoverishment. They resemble those observed in the course of male-pattern balding, although their development is less marked.

Based on earlier studies carried out by Saitoh *et al.*[1] on the growth cycles of body hair, in 1975–76 we developed a non-invasive method[2] of assessment of the principal parameters of hair growth and loss, and of the action of products capable of affecting these phenomena.

Previous studies describing modifications of follicular structures linked to hair cycles,[3–5] as well as general data on scalp hair or on alopecia,[6–8] have involved invasive methodology (biopsy, trichogram), and were not appropriate for observation of the succession and duration of hair cycles.

The results obtained during our early observations[2] led us to extend the study of hair cycles, not only to elicit interindividual variations, but also to compare individual hairs.

Our study group consisted of members of our laboratory personnel. There were 10 male volunteers, with or without alopecia, in whom we carried out photographic studies of the same scalp area, each month for 8–14 years, allowing us to monitor the progress of some 9000 hair cycles in all. The exceptionally long duration of this study allows us to describe the ageing process observed in these 10 subjects through the evolution of hair cycles, hair renewal capability, and hair diameters.

These phenomena have been likened to those observed in balding subjects, alopecia being sometimes considered a premature ageing phenomenon.

This work is a descriptive study, with the objective of expanding the knowledge and understanding of the phenomena linked to hair cycles.

Methods

Technique

The photographic technique, developed from that described by Saitoh[1,2] for the observation of body hair, is based on the following principle: after shaving an area of the scalp, it is possible to distinguish between the hairs in anagen, which lengthen by about 0·35 mm/day, and the resting hairs in telogen.

The camera used is equipped with a lens for macrophotography (enlargement × 2) and a ring flash. The camera is mounted on a rigid frame which serves to establish a fixed distance between lens and scalp area. The section of the frame in contact with the scalp contains a window equipped with a glass slide which covers the field of view, and flattens the hairs on to the scalp surface.

An area of the scalp containing about 80 hairs was accurately delineated and shaved. A photograph of the area, taken 2 or 3 days after shaving, was examined to determine the current phase of each hair (Fig. 1): anagen, if hair growth was observed, telogen, if no growth was apparent, and to measure the diameter of each hair, by comparison with a series of calibrated threads (μm): 1 very fine < 35; 2 fine 35–50; 3 medium 51–65; 4 thick 66–80; 5 very thick > 80.

Hair distribution within a single area (clustering, direction) was found to be highly characteristic, and hence two photographs of the same area, taken at different times, could be superimposed, and each hair or cluster of hairs made to coincide (Fig. 2). Each hair

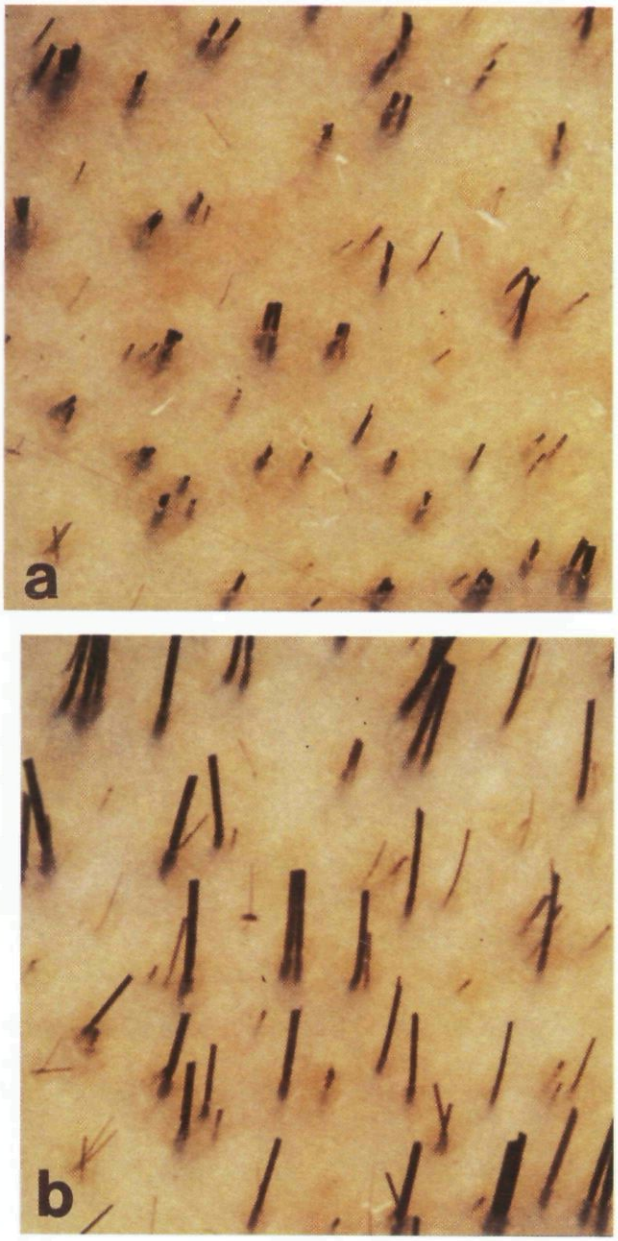

Figure 1. Phototrichograms: principle. (a) D_0: hairs shaved on an area measuring approximately $0{\cdot}5\,cm^2$. (b) D_2: two days later, hairs in anagen have grown approximately 1 mm, but telogen hairs show no growth.

could thus be identified, and its behaviour monitored over the course of time.

Notes

1 In the course of such a prolonged study, it is almost impossible to impose a strict hair-washing timetable on the volunteers. In order to make the sites examined as uniform as possible, they were cleansed using a cotton-wool bud soaked in 90% alcohol before each photograph, to remove any debris such as dirt particles, hair fragments or scales.

2 The study site on the scalp was established by a system of right-angled coordinates. An instrument based on the principles used in neurosurgery was developed for this purpose.

Volunteers

Ten male volunteers were studied over a period of 8–14 years (Table 1). With the exception of volunteer H, aged 49 years, all were between 25 and 32 years old at the start of the study. Their hair colour was light to dark brown. On the scalp of each volunteer, an area situated on the vertex, between the frontal region and crown, was selected for study. This region was chosen because male-pattern baldness initially affects the frontal region and crown, and then subsequently the intermediate vertex; the latter is thus the region of choice for charting the progress of alopecia. A phototrichogram was recorded every 4–6 weeks.

The volunteers were allowed to use cosmetic hair cleansing products of their own choice. None used a preparation known for its action on hair loss (minoxidil).

Hair density was greatest in volunteers A, B, C and D, who had no sign of alopecia, and the proportion of hairs in telogen amounted to less than 15% in each of these cases.

Volunteers E, F, G, H, J and K showed obvious alopecia: E and J with only frontotemporal recession (grade III on the Hamilton scale[9])—the proportion of their hair in telogen approached 20%; F, G, H and K manifested more prominent frontal recession and thinning hair on the vertex, and fell within grades III–V on the Hamilton scale. The proportion of hair in telogen was in the region of 30%. In all six volunteers with alopecia, overall hair density was relatively low.

Data collection and processing

In each of the 931 hairs identified, the successive phases of the cycles were observed: anagen, telogen and the latency period. 'Latency period' is defined as the interval between elimination of a hair in telogen and the appearance of the replacement hair in anagen, beginning the next cycle. During the latency period the hair site remains unoccupied. It was observed to occur in 80% of hair cycles. Observations were made at intervals

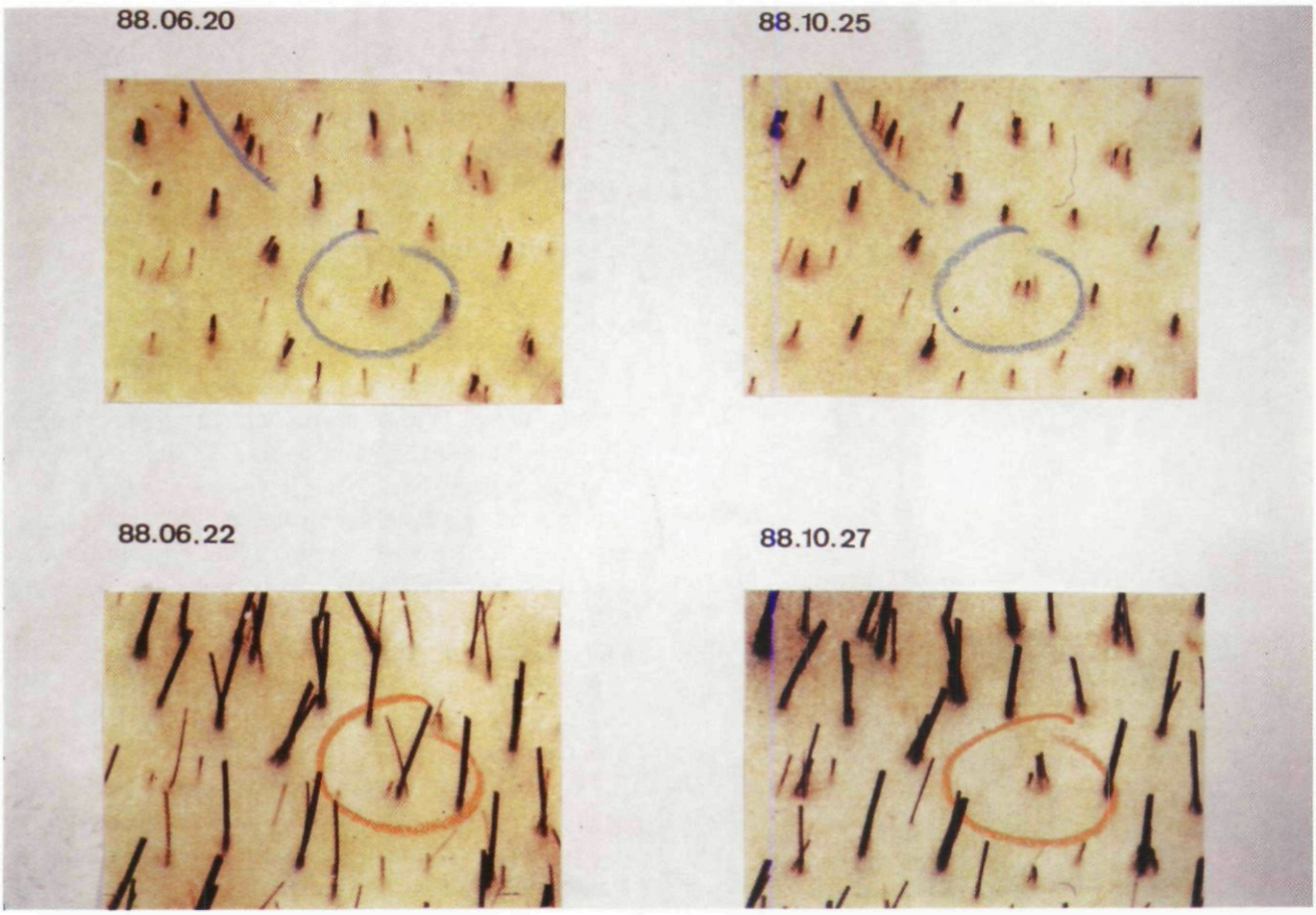

Figure 2. Phototrichogram of a single scalp area over the course of time. Left: 20 and 22 June. Right: 25 and 27 October.

of approximately 1 month, and the duration of a phase was thus expressed in months.

The results were stored and analysed with the help of computer programs (RS1, SAS, BMDP) within the VAX system. The statistical analyses carried out were intended to furnish a better explanation and understanding of the phenomena linked to ageing of scalp hair.

Table 1. Characteristics of the 10 male volunteers

Subjects	Age at the start of the study	Hamilton classification (modified Norwood)	Duration of study (months)	Hair count/area studied	Hair characteristics at the beginning of the study (1st year)	
					Hairs/cm^2	Anagen hairs (%)
A	32	I	143	94	283	92
B	25	I	162	81	289	90
C	26	II	157	84	313	85
D	28	II	162	92	269	86
E	25	III	129	144	196	82
F	32	III vertex	162	77	261	65
G	30	IV a	114	90	272	69
H	49	V	119	48	197	69
J	25	III	95	69	179	77
K	29	III vertex	126	62	215	64

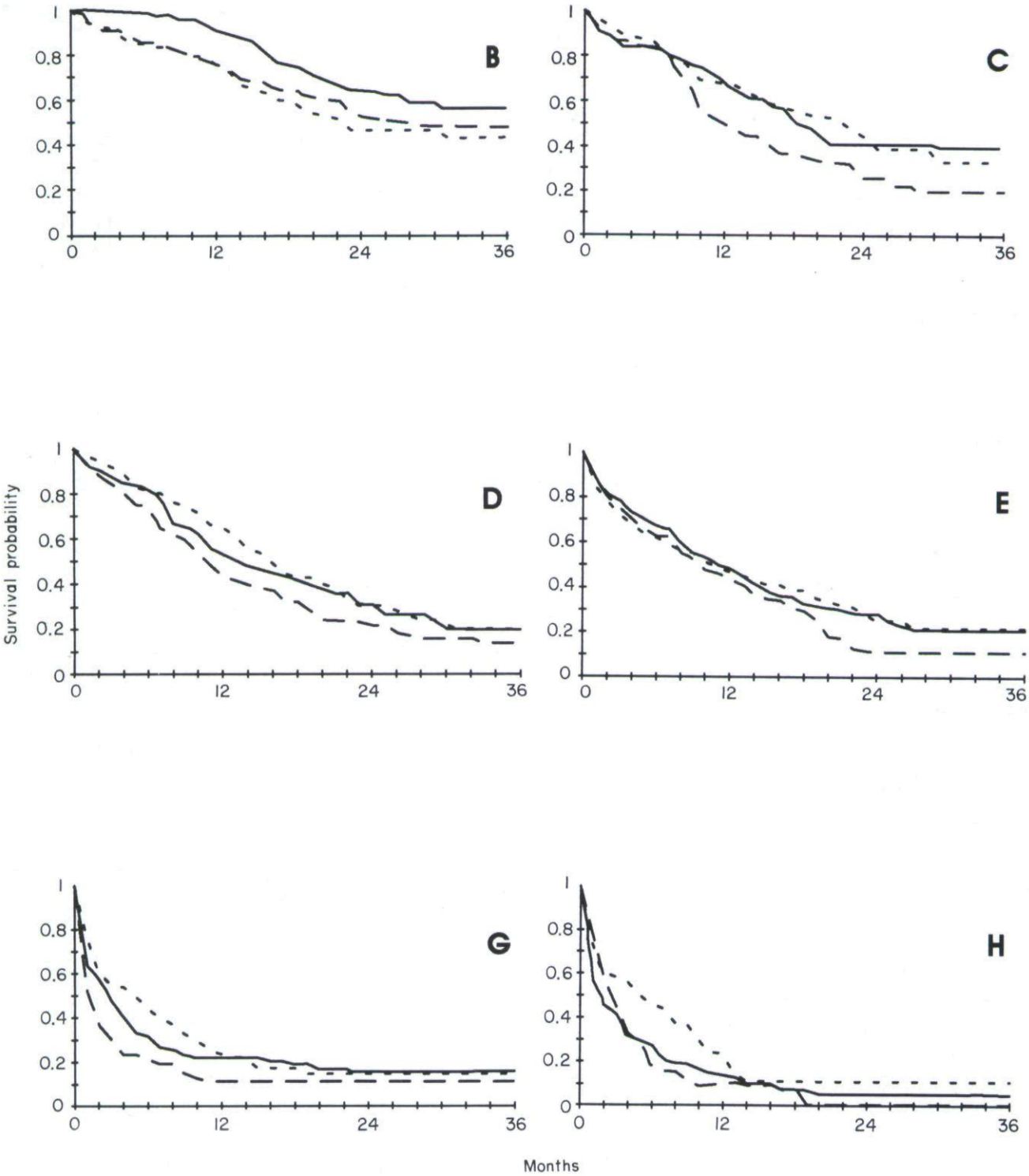

Figure 3. Evolution in lifespan of hair. Each curve represents, for the hair of a single subject during the 3-year period under consideration, 'the survival probability', defined as the percentage of hairs whose growth duration is greater than x months (x on the abscissa). - - - -, period I; ——, period II; — —, period III.

Statistics

Analysis of variance (ANOVA) was used to assess the significance of observed differences, the significance level being $P < 0{\cdot}05$. It took the form of an overall analysis, taking into account simultaneously: (i) the duration of each individual phase (anagen, telogen, latency); (ii) the related diameter during each of these phases; (iii) ageing, evaluated in terms of a comparison between first and last cycles.

Results

Evolution in lifespan of hair

The 'lifespan of hair' relates to the duration of the anagen phase. Although telogen forms part of the whole lifespan, we have shown [10] that, throughout the whole group of cycles observed in 10 subjects (7727 complete cycles), the telogen phase is short, of more or less constant duration, and comparable for all the cycles (2–3 months).

The percentage of hairs whose observed period of growth was equal to or greater than x months (x varying from 0 to 36 months) was calculated for each individual at regular intervals throughout the course of the study. Three periods were chosen for these calculations: i, 1977–80 (or 1975–78 depending on the individual); ii, 1981–84 (or 1979–82); iii, 1985–88 (or 1983–86). This percentage is defined as the 'survival probability'.

The curves obtained (Fig. 3) demonstrate that the ageing process in these male volunteers was evinced by a general reduction in the lifespan of a hair. However, this phenomenon was manifested to different degrees in different individuals.

Classified according to lifespan, not all classes of hair were necessarily affected to the same degree. For example, in subject E, a reduction in lifespan, over the course of the study, was only observed in those hairs which had previously survived for longer than 18 months, whereas in subject D the reduction appeared homogeneous throughout all classes of hair lifespan.

Periods of marked change alternated with periods of relative stability. In subject C, for example, the evolution was more marked at the end of the study (period III compared with period II) than at the beginning (period II compared with period I). In subject B, mean lifespan decreased during period III compared with II, but in the same subject the lifespan had previously been seen to increase during period II compared with I.

Table 2. Overall analysis of variance. Study of anagen duration in each subject relative to hair diameter [category 1 (very fine) to 5 (very thick)], in terms of the ageing factor; anagen in the earliest (A1) and latest (An) cycles

Subject	Ageing An vs. A1	Category of diameters affected ($P < 0{\cdot}05$) by shorter growth duration
A	An < A1 $P = 0{\cdot}01$	3 and 4
B	An < A1 $P = 0{\cdot}013$	3 and 4
C	An < A1 $P = 0{\cdot}028$	3 and 4
D	An < A1 $P = 0{\cdot}004$	3 and 4
E	An < A1 $P = 0{\cdot}0001$	4 and 5
F	An < A1 $P = 0{\cdot}006$	3
G	An < A1 $P = 0{\cdot}0001$	5
H	An < A1 $P = 0{\cdot}0016$	3
J	An < A1 $P = 0{\cdot}0006$	4 and 5
K	An < A1 $P = 0{\cdot}06$	3

This reduction in lifespan was confirmed by variance analysis: the duration of the anagen phases decreased significantly in the course of ageing in all subjects (Table 2).

Hair replacement

The telogen phase, the resting time necessary for elimination of a hair, did not appear to be influenced as the subject aged. However, the latency period lengthened in all individuals, except subject A. The degree of prolongation was significant in subjects B, C, F, G, H and K.

In subjects suffering from mild or no alopecia (A, B, D,

Table 3. Anagen, telogen and latency periods calculated after the first and the last cycle observed

	Anagen period (months)		Telogen period (months)		Latency period (months)	
Volunteer	First cycle	Last cycle	First cycle	Last cycle	First cycle	Last cycle
A	30·1	18	1·9	2·1	6·2	5·6
B	31	19·4	2·1	2·1	5·2	9·7
C	18·4	10·7	2·7	2·3	5·9	11·9
D	19·1	11·7	2·5	2·2	3·3	3·8
E	15·5	8·5	2·3	1·9	2·1	2·8
F	5·6	4·3	2·5	2·2	2·7	6·7
G	13·3	7·6	2·8	2·9	4·1	7·5
H	7·4	4·2	2·9	2·4	2·4	5·4
J	10·6	5·6	2·3	2	2·5	3·5
K	7·5	5	3	2	5·1	11·4

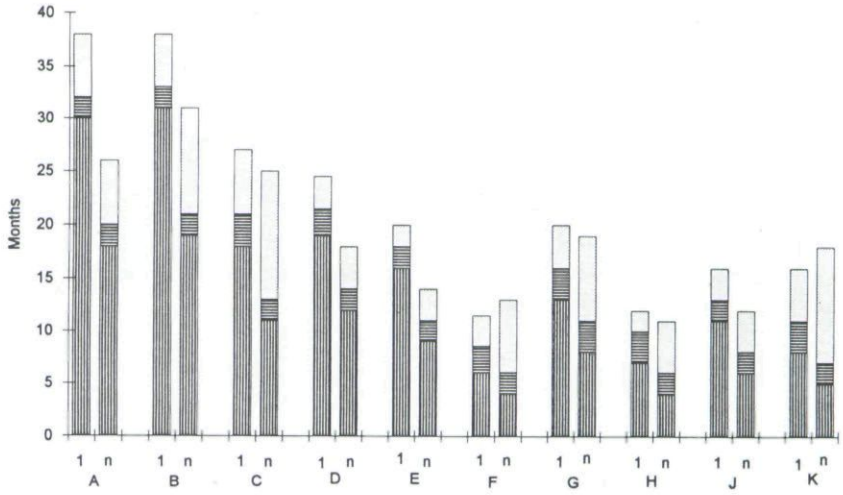

Figure 4. Phase durations during the first (1) and last cycles (n) observed in each subject. ▒, latency; ▤, telogen; □, anagen.

E and J), prolongation of the latency period was less marked than the reduction in the growth duration; there was a concomitant increase in the frequency of hair cycles.

In the subjects who had more obvious hair loss (F, G, H and K, but also in subject C), the short growth phases, which became even shorter in the course of ageing, and the prolongation of the latency period, are signs of an overall slowing down of activity by the hair follicles. As an illustration of this, Table 3 and Figure 4 show the mean values of the duration of phases in the first and last cycles observed.

Relationship of hair lifespan to diameter

In every subject, the finest hairs had the shortest growth phases and, in contrast, thicker hairs enjoyed longer periods of growth. Table 4 illustrates this correlation during the earliest cycles observed. The association was highly significant ($P < 0{\cdot}001$) in every case. It was maintained throughout the course of this study in all subjects, but the reduction in growth duration in the course of ageing affected primarily the hairs with the greatest diameter, leading to an overall reduction in hair diameter over the course of time.

A detailed analysis of the evolution in diameter of each hair revealed two contrasting phenomena: on the one hand, a tendency towards lower values in 38% of hairs, and on the other, a tendency towards higher values. The latter involved 16% of hairs, and was most evident in the subjects displaying the least degree of balding (A, B, D, E).

The effect of these findings on the overall figures (Fig. 5) for the evolution of hair diameter was that a reduction in hair diameter occurred ($P < 0{\cdot}01$; Student's *t*-test) during the course of ageing.

Table 4. Growth phase duration relative to hair diameter in each subject

	Mean duration of anagen during the first cycles observed, relative to hair diameter category 1 (very fine) to 5 (very thick)				
Subject	1	2	3	4	5
A	3·6	16·1	36·5	38·1	–
B	–	7·6	27·1	44·3	–
C	2·6	8·2	19·4	27·4	–
D	–	10·4	20·4	30·2	–
E	2·0	7·8	7·9	21·1	23·0
F	1·5	3·8	6·5	11·1	–
G	2·1	5·8	8·2	22	72·4
H	1·6	5·3	15·8	14	–
K	0·7	–	6·2	10·4	19·1
J	3·2	3·4	12·4	12·9	–

Discussion

In general, ageing was reflected by a decline in the activity of hair follicles. Growth duration was curtailed, the hair shaft diameter diminished, and the interval between two cycles extended. The consequences for scalp hair were:

1 A reduction in the number of hairs per unit area. Growth cycles being shorter, hairs were 'absent' more frequently, and for longer periods, as the time required for replacement was extended. This reduction in density (Table 5) was not identical in every case. It amounted to less than 10% in 10 years in the individuals with the least alopecia (A, B, D, E and J), but was much more

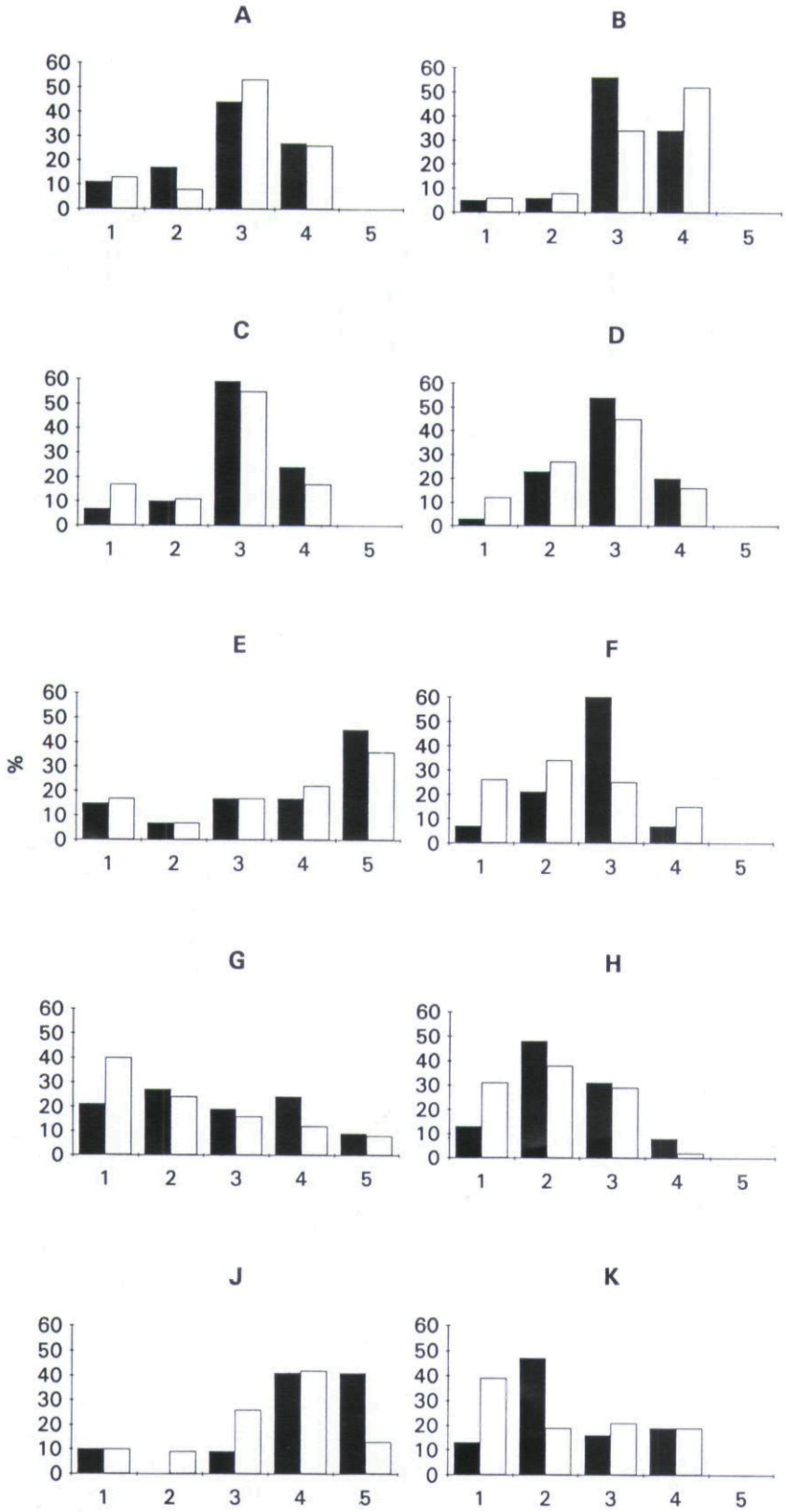

Figure 5. Comparison of the distribution of hair diameters on a scale ranging from 1 (very fine) to 5 (very thick), at the commencement (■) and end of the study (□), in each subject.

Table 5. Changes in hair density

Volunteer	Hair/cm^2 At the beginning of study	Hair/cm^2 At the end of study	Variation (%)	Observation time (months)	Yearly variation (%)
A	283	283	0	143	0
B	289	268	7·3	162	0·5
C	313	264	15·7	157	1·2
D	269	252	6·3	162	0·5
E	196	185	5·6	129	0·5
F	261	213	18·4	162	1·4
G	272	209	23·2	114	2·4
H	195	172	11·8	119	1·2
J	179	174	2·8	95	0·4
K	215	161	25	126	2·4

evident in the balding subjects F, G, H and K, as well as the non-balding subject C.

It should be borne in mind that these results concern observations and hair-counts carried out on zones midway between the frontal area and the crown.

On an individual scalp, the extent of the reduction depended upon the position of the zone studied; some areas particularly prone to alopecia (frontal area and crown) suffered a rapid decline in density over a period of a few years.

2 Deterioration in the quality of the scalp hair. Ageing engendered a significant reduction in growth duration. The maximal length attained by a hair therefore diminished as the subject aged. In parallel, the hairs became finer; hair quality thus declined.

It should, however, be noted that, although a reduction in hair diameter was the dominant tendency in the ageing process, there was also a tendency among non-balding subjects for the proportion of thicker hairs to increase; this tendency was noted by Barman *et al.*,[8] who detected an increased proportion of thick hairs in their oldest subjects. One explanation for this might be that it merely reflects the more frequent and longer absence of the finest hairs.

These two components of ageing, quantitative and qualitative, combine to impoverish the scalp hair.

Ageing did not appear to follow a perfectly regular course over time. Periods of stability, or even partial remission, alternated with periods of more marked evolution. Individual factors such as the subject's general health and life-style may well play an important role in this evolution.

The processes implicated in ageing, in terms of the succession of hair cycles, appear to have consequences similar to those implicated in male-pattern baldness. Alopecia, characterized by rapid and visible impoverishment of scalp hair, gives the appearance of accelerated ageing. Ageing, in turn, may impose on elderly subjects a head of hair resembling that of the early stages of alopecia. However, this is not to say that the biological mechanisms regulating the two phenomena are necessarily one and the same.

The attempt to combat ageing of scalp hair, and the search for 'anti-hair loss' or 're-growth' treatments are thus challenges with a great deal in common. Better understanding of the mechanisms brought into play in the stimulation of follicular activity, and in the triggering of new hair cycles, should result in improvements to existing methods of treatment, and should elicit novel solutions to the problems of hair deficiency.

Acknowledgments

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